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Task Description:	Maintaining good health during long term space travel requires that all water, potable or hygiene, must be continuously monitored. Ion chromatography (IC) is the best method for monitoring 23 of the 37 monitored contaminants, but conventional IC equipment is heavy (~38 kg), bulky, and requires a constant supply of reagents. Both Lynntech, Inc., and NASA's Jet Propulsion Laboratory (JPL) have been developing miniaturized IC analytical instruments that are much lighter, smaller and nearly reagent free. (The Lynntech instrument only requires deionized (DI) water a small supply of potassium sulfate while the JPL instrument only requires DI water.) Although they have significant differences, both instruments are capable of carrying out determinations on all of the ions normally determined with a full scale IC instrument to levels as low as a full scale IC instrument. The Lynntech instrument has demonstrated the detection and quantification of alkali ammonium ions at the ppm level. The JPL instrument promises rapid (5 min)		

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analyses at all levels. The technologies behind both instruments are fully compatible with microgravity operation and the best features of both devices will be combined into a single instrument. The instrument to be built in this project will be fully self-contained, with closed fluid handling and complete self-calibration capabilities, and will be fabricated with microgravity operation in mind. In this project both technologies will be advanced and tested against common benchmarks. This instrument will be fabricated and tested in the AWRS testbed at the Johnson Space Center. The goal of this project is to produce a flight test ready device.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Microfluidic Ion Chromatograph for In-Flight Water Quality Analysis. Tony Ragucci, Francisco Maldonado, Marius Raducanu, and Alan Cisar Lynntech, Inc.

ABSTRACT. Although water quality may currently be analyzed on the ground after a flight, long-duration missions will require the capability to perform analyses on-board. If a water purifier fails, contaminants must be detected rapidly and corrective action taken in a timely manner to prevent serious harm to the crew. Many of the possible contaminants which could negatively affect astronaut health are inorganic ions. These ions can be quantified by ion chromatography (IC), although current commercially-available IC's are too large, heavy, and power-intensive to be used on a space mission. These units also require large quantities of caustic chemicals for analysis, which would pose a significant hazard in a microgravity environment. To meet the need for an inorganic water quality analysis device for long-duration missions, Lynntech developed an ion chromatograph tailored for future planned long-duration missions. The entire unit is the size of a shoebox, requires little power, produces anionic and cationic chromatograms simultaneously, and generates the acid and base needed for analysis from benign salt water. Several components were developed specifically for this application over the course of work. A developed microfluidic liquid/gas phase separator performed successfully in microgravity conditions; a syringe pump was developed to provide uniform and repeatable high-pressure flow from a low-power, compact device; and a conductivity detector was designed and implemented with six orders of magnitude in dynamic range, an internal volume of only 0.7 microliters, and an autoranging capability which allows the analysis of both high and low ionic species concentrations in the same sample. The complete system and its measurement capabilities are described in detail.

INTRODUCTION. The need for clean regenerated water on planned long-duration missions to the moon and Mars is clear. Whereas current shuttle and ISS missions rely on water resupply to support astronauts, this scheme becomes excessively onerous for relatively remote missions extending over hundreds of days. Assuming even the most conservative numbers1 for steady-state consumption rate (2.67 kg/CM-d), crew size (4), and Mars mission duration (657 days); over 7,000 kg (7.7 tons) of fresh water would be required to sustain the crew without a water regeneration system. If water regeneration is to be used, then the need for water quality analysis will be critical.

Recycling all water introduces the potential for crew poisoning if the wastewater contamination levels are not reduced to permissible levels. Onboard water quality monitors are required to frequently analyze the crew water supply in order to quickly indicate if a water purification unit is not functioning properly. In broad terms, the quality of on-board water can be broken into two aspects: organic and inorganic contamination. Overall organic contaminant loading can be analyzed through total organic carbon analysis, as described in another article co-published through this ICES conference. Inorganic contamination analysis, however, requires more specific identification, such as is possible with ion chromatography.

In October of 2003, a NASA-sponsored panel of water quality experts from within and external to NASA met to review state-of-the-art developments in water quality analysis for future ISS missions.2 As part of that evaluation process, a preliminary list of detection targets for inorganic contaminants was produced. The primary targets were ammonia, iodine, lead, nickel, and silver, since these are both hazardous in trace quantities and commonly found onboard as biocides and coolant, and through sources such as stainless steel plumbing and recovered humidity condensate. These contaminants can be readily analyzed through ion chromatography, as they often are terrestrially.

Extensive research has been ongoing to develop a microchip-level, lab-on-a-chip liquid chromatograph and significant successes have been achieved in this area.3,4,5,6 However, the chip-based platforms generally require lab-scale components for some aspect of the analysis such as interfacing, pumping, or detection, which substantially increase the overall system mass, volume, and power demands. Additionally, none of these systems can be reused more than a few times. Therefore, many disposable chips would be needed for a long-duration mission. There currently is no commercial system available which can meet the requirements for size, weight, power requirements, and microgravity compatibility to analyze water on-board.

Lynntech has chosen an alternate approach based on the miniaturization of existing components while retaining the proven reliability of a traditional system. Standard separation column / ion suppression / conductivity detection analysis is performed. However, the reagents needed for analysis are generated electrochemically from a benign salt, reducing the primary hazard associated with using the device in a space environment. Additionally, the system has been miniaturized roughly to the size of a shoebox. In the original design of the Ion Analyzer, the pumping system used to drive eluent through the separation columns was electrochemical in nature. However, these pumps were replaced with specially-designed piston pumps to ensure reliability and accuracy over a multiple-year timeframe. This was done in such a way as to not increase the overall Equivalent System Mass (ESM), significantly.

Whereas previous efforts focused on proof-of-concept7 and preliminary breadboard8 demonstration of the IC, the goals of more recent work have been the engineering aspects of further miniaturization, robust system design, and ease of operation in addition to improved stability and measurement performance. This new system has now been designed and built and the demonstrated operating parameters are significantly lower than that of comparable commercial systems.

SYSTEM DESIGN. The IA operates at a basic level in the same way that a bench-scale ion chromatograph does. However, generation of acid and base from saltwater, minimization of system pressure by using short chromatography columns, and overall reduction in size, weight, and power requirements does change the system in some respects from a typical bench-scale IC. The diagram in Figure 1 indicates the overall interoperation of the system components. Sample is injected at an inlet port from a sample syringe, and liquid sample flows through two 100 µL injection ports for separate

cation and anion analysis. Potassium sulfate saltwater is drawn by a peristaltic pump into the acid/base generator, which electrochemically matches the dissolved salt ions with hydrogen and hydroxide from water to form the sulfuric acid and potassium hydroxide needed for cation and anion analysis. The acid and base are stored in small bladders and then drawn, when needed, into syringe pumps. The pumps then force the carrier solution through the injection valves, once they have been actuated by the user, and drive the sample into both separation columns. On exiting the columns, the acid or base in the eluent stream is removed electrochemically to reduce the background conductivity of the solution. Finally, the ions are detected by conductivity detectors, which measure both retention time and analyte concentration through integrated peak area.

MAIN COMPONENTS. Development of the Ion Analyzer required the complete design of each component used in the system, with few exceptions such as the Dionex chromatography columns used. Operation, design and testing of each of the main system components are described in detail below.

CONDUCTIVITY DETECTOR. Two conductivity detectors are used in the instrument, enabling simultaneous cation and anion analysis. The conductivity detectors must have a wide measurement range so that large concentration peaks can be resolved in addition to trace contaminants. In order to meet this need, a detector was developed with six orders of magnitude in dynamic range (0.1 µS – 10 ms) and a detection cell volume of only 0.7 microliters to reduce band-broadening. The conductivity detector also has an autoranging capability which allows the analysis of both high and low ionic species concentrations in the same sample. If the sample composition is completely unknown, an autoranging test run can be performed to determine the optimal range setting to use for a subsequent, high-resolution analysis. The conductivity sensor system consists of (i) a foam block holding the conductivity cell, heaters, and a thermistor, (ii) electronic mother and daughter boards mounted under a standoff plate, and (iii) a welded aluminum housing fitting over this assembly, creating an RF-tight enclosure. The heater block and insulating foam provide stable thermal isolation and control. Polyimide (Kapton®) tape heaters on the block are capable of maintaining the temperature of the core cell within 0.1°C and also heat the sample water routed through the inlet and outlet tubes. When assembled, the aluminum blocks encase the cell to provide high thermal mass; the light-density, tight-fitting, PVC foam minimizes heat loss through convection. The large thermal stability provided by this design minimizes conductivity drift produced by ambient thermal fluctuations. By enclosing the entire system in a water-tight aluminum housing, electromagnetic interference which might affect the system is substantially reduced. The system is, therefore, more capable of withstanding external electromagnetic noise.

Repeatability of the conductivity sensor system was tested by injecting a series of 100 µL-volume conductivity standards (KCl in water) directly into the conductivity cell using a loop valve.

ACID/BASE GENERATOR. The acid/base generator provides the eluent needed for analysis of both cations and anions. A solution of K2SO4 is injected into the center of three thin channels in a monolithic polyvinylidene fluoride (PVDF) block. The center channel is separated from a channel above and a channel below by anion and cation exchange membranes. An electric field produced between electrodes in the two outer channels drives the sulfate and potassium ions across the ion exchange membranes where they combine with hydrogen and hydroxide ions in the outer channels to produce acid and base. The new design for the eluent generator, as well as for the ion suppressors, is more compact, rigid, and chemically inert than its predecessor made in PDMS. The unit uses standard HPLC fittings and screws, so that it can be taken apart and reassembled, if necessary, using only a single screwdriver.

The output of the acid / base generator is well controlled by the voltage applied to the unit. A series of three tests were performed. In each test, a solution of 0.115 M K2SO4 was pumped into the unit at 0.25 ml/min while water was pumped into the outer channels at the same rate. Power was applied at 1 V (2 mA), 2 V (9 mA), and 3V (25 mA) and the pH of the generated acid and base was measured. The performance consistency enables the generation of controlled concentrations of acid and base for many types of analysis.

DUAL SYRINGE PUMP. Much of the mass and power requirements for a traditional IC are due to the syringe pumps required to drive eluent through separation columns. A miniature dual syringe pump was developed to enable simultaneous cation and anion analysis with only a fraction of a traditional system's mass and power requirements. The pumps deliver acid or base at up to 200 psi in a single piston stroke, sufficient for analysis on a short column. The delivery rate is regulated to ensure smooth flow over 30 minute chromatographic separations. Although limiting the run duration to 30 minutes, the use of a single syringe eliminates the need for pulse dampeners or flow restrictors. Basic testing demonstrated that the flow delivery is uniform enough that no steps in conductivity were detected during peak detection.

PHASE SEPARATORS. The IC prototype includes multiple mesofluidic phase separators designed to remove any bubbles from the liquid flow. It is critical to remove all bubbles from the sample stream to prevent, for instance, the instrument from generating an erroneous result due to a bubble passing through a conductivity sensor.

The phase separator uses an expanded polytetrafluoroethylene (ePTFE) tube, bent into a small spiral configuration. The ePTFE tubing has a pore structure which allows gas to pass through its wall but restricts water from passing through. Because of internal fluid pressure, the small inner diameter of the tubing, and a spiral flow path, gas bubbles are driven towards the ePTFE wall and out of the tubing. The dimensions of the phase separator are 5.4 cm x 2.0 cm diameter, it weighs 11.7 grams, and it is completely passive.

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Phase separation is a system-critical process which is typically most sensitive to the influence of gravity. Therefore, the mesofluidic phase separators were tested, alone, in a zero-g environment aboard a C-9B aircraft under various gas and liquid flow rates. Two independent peristaltic pumps drew from a liquid water and nitrogen gas reservoir. The two fluids were then mixed and injected into the phase separator. The exit streams from both the liquid and gas ports were continuously monitored with high-resolution video cameras to observe any bubbles 10 picoliters or larger in the liquid stream or liquid exiting the gas port. Additionally, the effective X, Y, and Z -gravitational acceleration, temperature, absolute pressure, and relative humidity were recorded to correlate video images with environmental conditions. No failures were detected during normal (1-g) flight. Gas which did exit the liquid port of the phase separator was detected by computer-aided visual analysis of the data. The mesofluidic phase separator was highly effective at removing bubbles in zero-g conditions. The liquid flow rates for all sections of the Ion Analyzer are set to 0.25 ml/min. Therefore, a gaseous volumetric flow rate of up to 10 times the liquid flow rate can be reliably removed in microgravity conditions.

ION SUPPRESSORS. The baseline conductivity of the eluent stream coming from the separation columns is typically several millisiemen, due to the high conductivity of the acid or base used as the mobile phase. A peak height of only 1 µS for a sample concentration on the order of 1 ppm, for instance, would be difficult to measure on such a high

background. Therefore, the ion suppressors remove the acid or base from the eluent stream while leaving the ions of interest in the liquid stream to be quantified by the conductivity detectors.

The selective removal of acid or base is done electrochemically in much the same way that the acid/base generator originally removes sulfate and potassium ions from a center channel to outer channels to produce the acid and base. An electric field is generated between electrodes in the two outer channels and the sulfate (potassium) ions are driven across a selectively permeable membrane to combine with hydrogen (hydroxide) ions from water in an outer channel and re-form acid (base) external to the original flow path.

Water flows in the outer channels, as introduced through HPLC fittings on the top and bottom of the device. The sample stream flows directly through the center channel without meeting flow obstructions from either of the two electrodes to disrupt the laminar flow or induce mixing. Electrical connection is made through separate HPLC fittings on the top and bottom of the unit. Testing of both ion suppressors was performed using a stock solution of known conductivity and flowing it through the center channel of the suppressor at 250 µL/min while simultaneously flowing water through the outer channels at a rate of 250 µL/min each. A solution of 5.9 mS KOH was used for the anion suppressor and 5.75 mS H2SO4 was used for the cation suppressor. Current control was applied and the variation in voltage was recorded. Applying more than 25 mA to either of the two suppressors was sufficient to reduce the center channel outlet conductivity to less than 10 µS.

PERISTALTIC PUMPS. Miniature peristaltic pumps drive fluid throughout the instrument smoothly and also provide an automatic check against counter-flow when inactive. However, the tubing longevity typically limits the pump lifetime for a maintenance-free instrument.

W. L. Gore & Associates developed a new peristaltic tubing formulation based on a composite of platinum-cured silicone and ePTFE which extends the operational tubing life to 3,000 hours. Although Chem-Sure® tubing is not yet widely available, Lynntech established a collaboration between Instech Laboratories, and W. L. Gore & Associates to modify a current Instech pump design to use the new tubing. The projected tubing life now matches the expected lifetime of the pump mechanism, extending the total operational lifetime of the pumping system to allow over 6,000 IC analyses to be performed. Although these pumps were not installed on the system at the time data shown was collected, separate testing has demonstrated that the new tubing performs equivalently to the common peristaltic tubing then in use.

HANDHELD PC INTERFACE. A handheld computer is used both to control the system and provide graphical and numerical feedback on analysis progress and results. Software was developed for the Windows CE operating system, as shown in the main menu and control panel screen shots in Figure 8. The system can run an analysis with the conductivity detector in autoscale mode, or a static scale can be used for greater accuracy. All data is automatically logged and time-stamped. The unit is also detachable so that data analysis can be performed away from the machine after a chromatograph has been collected. To begin an analysis, the user inputs the desired flow rates and currents for the mobile phase pH and flow in the Control Panel, or accepts the defaults, and selects the conductivity scale to use during analysis. An autoscale option is available to roughly quantify multiple sample ions, which may vary significantly in concentration. However, due to discontinuities present at the scale changes, precise measurement in this mode is currently not possible. After establishing the concentration magnitude for the targets of interest, or if the concentration is approximately known, a static scale (e.g. 10 µS/cm) can be selected for a sample run.

The user then returns to the main menu and starts either the cation or anion pump or, typically, both simultaneously. Data streams from the conductivity detectors to the handheld PC where it is automatically logged and time-stamped in a new file. The user then injects his sample into the sample loop from a syringe and actuates the injection valve(s). From the live graph, the incoming data can be monitored in real time or the analysis can run autonomously. The maximum analysis time using the standard system flow rate of 250 µL/min is 30 minutes. If an analyte requires more time to elute from the separation column than this, a more concentrated eluent must be produced from the acid/base generator to reduce the retention time. After the data collection is complete, the current or previously recorded data can be analyzed. From the Analysis Panel, the user inputs values for the measurement baseline; peak threshold, corresponding to the relative trigger level for detection of a new peak; area threshold, to eliminate spurious peaks due to simple noise; and a maximum peak detection number, to prevent the program from running for a long time if the analysis parameters were input incorrectly. The program uses this input to automatically detect peaks in the chromatograph, determine analyte retention time for ion identification, integrate the area under the curve for each peak, and compute the fractional concentration of that ion relative to the whole sample. Although not currently implemented, the program could also cross-check data for a given eluent pH with a small internal database of known probable contaminants to directly identify ions from peak retention times.

PROTOTYPE DEVELOPMENT. Combining all of the aforementioned system components, the Ion Analyzer is completely self contained. All reagents, fluid manipulation, detection, control electronics, and user interface hardware and software is located within the enclosure. Only 25 W of power are required to operate the instrument.

For testing, a mixture of NaCl, KCl, and CaCl2·2H2O (Sigma Aldrich) was prepared from stock solutions which were serially diluted to produce 2 ppm Na+, 5 ppm K+, and 5 ppm Ca+2. Using the default settings, an automated run was performed with Dionex IonPac columns AG14 and CG12A and analyzed.

CONCLUSION. The Ion Analyzer has demonstrated potential for performing inorganic water quality analysis on future long-duration space missions. It is considerably smaller, lighter, less power consuming, and safer than existing commercial systems for performing ion chromatography. Microgravity operability testing has begun through validation of the phase separators developed for this application, which have performed successfully in zero-g flight tests. Toxicological hazards typically associated with performing liquid chromatography have been substantially reduced, since consistent acid and base generation eliminates the need to bring caustic reagents on-board. The conductivity detector developed provides the measurement range, consistency, and linearity required for comprehensive water analysis.

This device could provide critical water analysis capability on future long-duration missions for inorganic contaminants of concern. Additionally, the system could be coupled with an organic counterpart, such as a total organic carbon analyzer, to provide comprehensive water quality analysis. Many components are in common between these systems under development and, as such, they could be combined to provide one complete water quality analysis instrument.

Clearly, more work is required to generate ion chromatograms of a quality matching bench-scale state-of-the-art IC systems. However, the demonstrated potential is certainly clear. Further development work would concentrate on

improving baseline stability for accurate quantitative analysis, first, and increasing peak resolution (i.e. number of plates), second. Parallel work by others5 in the latter area indicate real potential for significant improvement. If realized, the technology would undoubtedly be an asset on future missions.

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CONTACT Lynntech, Inc., 1313 Research Parkway, College Station, TX 77845. E-mail: tony.ragucci@lynntech.com phone: (979) 693-0017 fax: (979) 694-8536. DEFINITIONS, ACRONYMS, ABBREVIATIONS COTS: Commercial off the shelf DI: Deionized ePTFE: Expanded Polytetrafluoroethylene ESM: Equivalent System Mass HPLC: High Performance Liquid Chromatography IA: Ion Analyzer / Ion Analysis IC: Ion Chromatograph / Ion Chromatography ISS: International Space Station PDMS: Polydimethylsiloxane PVDF: Polyvinylidene Fluoride RTD: Resistance Temperature Detector

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Papers from Meeting Proceedings	Ragucci T, Maldonado F, Raducanu M, Cisar A. "Microfluidic ion chromatograph for in-flight water quality analysis." To be presented at 37th International Conference on Environmental Systems, Chicago, IL, July 2007. SAE paper number to be assigned July 2007. , Jul-2007